Line sensor

DETECT | RASPBERRY PI | INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

An infrared LED shines at the floor and a detector measures how much light bounces back.

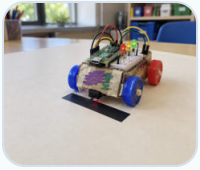
Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses



Warehouse guide-line robot

DETECT Read guide-line position → **DECIDE** Is the vehicle drifting? → **DO** Correct the steering



Cardboard lap-counting car

DETECT Cross dark finish strip → **DECIDE** Is this a new lap? → **DO** Increase the lap count



Infrared pair
Faces the floor

Adjuster
Sets the threshold

S
Digital signal

Raspberry Pi Foundation

Code

Raspberry Pi

```
from gpiozero import LineSensor

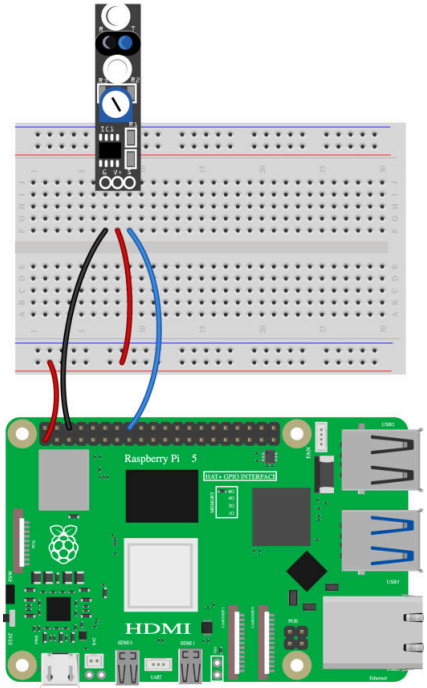
line = LineSensor(23)

on_line = line.is_active
```

> Turn the blue adjuster until the module's output changes cleanly at the edge of the line.

Wiring

RASPBERRY PI 5



GND (pin 6) → G, **3V3** (pin 1) → V+, **GPIO 23** (pin 16) → S